Jie Fu 傅杰

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Personal Statement

I am Jie Fu (傅杰 in Chinese). I am now a PhD student at Qiuzhen College, Tsinghua University.

Research Interests

Algebraic geometry and related areas, with current work on automorphism groups of hypersurfaces and hyperkähler manifolds. Further interests include moduli theory, rationality problems and derived categories.

Education

2025–present	**Tsinghua University**, Qiuzhen College, Beijing, China. Ph.D. student in Mathematics. Advisor: Prof. Zhiwei Zheng.
2021–2025	**Tsinghua University**, Qiuzhen College, Beijing, China. Bachelor in Mathematics and Applied Mathematics.

Publications and Preprints

Preprints

[1] **Remarks on Lattice Actions Corresponding to Automorphism Groups of Smooth Cubic Fourfolds**.
arXiv preprint, arXiv:2610.19161 [math.AG], 2026. 21 pages.
arxiv.org/abs/2610.19161

[2] **A Sharp Bound for the Second Betti Number of Hyperkähler Sixfolds**.
With Bin Wang, Shihao Wang, and Zhiwei Zheng. *arXiv preprint*, arXiv:2610.15803 [math.AG], 2026. 19 pages.
arxiv.org/abs/2610.15803

[3] **A Proof of Mongardi's Conjecture on Finite Symplectic Group Actions on Hyperkähler Manifolds of $K3^{[n]}$ Type**.
With Shihao Wang and Zhiwei Zheng. *arXiv preprint*, arXiv:2610.11109 [math.AG], 2026. 16 pages.
arxiv.org/abs/2610.11109

[4] **Classification of Automorphism Groups of Smooth Cubic Threefolds and Fourfolds**.
With Shihao Wang and Zhiwei Zheng. *arXiv preprint*, arXiv:2609.20403 [math.AG], 2026. 86 pages.
arxiv.org/abs/2609.20403

[5] **Automorphism Groups of Smooth Cubic Fourfolds through Lattice Theory**.
With Zhiwei Zheng. *arXiv preprint*, arXiv:2609.06683 [math.AG], 2026. 26 pages.
arxiv.org/abs/2609.06683

[6] **Non-symplectic Indices of Automorphism Groups of Smooth Cubic Fourfolds**.
With Shihao Wang and Zhiwei Zheng. *arXiv preprint*, arXiv:2606.11754 [math.AG], 2026. 30 pages.
arxiv.org/abs/2606.11754

Talks, Seminars and Academic Activities

- *Smooth hypersurface with automorphism group of large order*. Spring Learning Seminar in Algebraic Geometry, YMSC, Beijing. March 18th, 2026.
- *New results on automorphism groups of cubic fourfolds*. YMSC Seminar on Moduli Spaces and Related Topics, YMSC, Beijing. March 12th, 2026.
- *Symplectic automorphism of $K3^{[n]}$-type manifolds*. Hyperkähler Seminar, YMSC, Beijing. December 2nd, 2025.
- *Linear system of K3 surfaces*. Hyperkähler Seminar, YMSC, Beijing. October 24th, 2025.
- *Basic lattice theory following Nikulin*. Hyperkähler Seminar, YMSC, Beijing. September 30th, 2025.
- Co-organization: Student Seminar on Perverse Sheaves and Application to Representation Theory, Qiuzhen College, Beijing, Fall 2025.

Teaching Experience

Spring 2026	Teaching assistant, *Algebraic Variety* given by Prof. Junpeng Jiao, Tsinghua University.

Honors and Awards

Spring 2026	Pi Day's best problems award, Qiuzhen College.
Spring 2023	Honorable Mention, Analysis and Differential Equations, S.-T. Yau College Student Mathematics Contest.
Fall 2020	First prize in China's national high school mathematics competition, Chinese Mathematical Society.

References

Prof. Zhiwei Zheng
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Assistant Professor,
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